Introduction

In week 2, we calculated several physical properties, though in each case we assumed porosity was negligible. However, for many rocks porosity

Consider a sandstone composed of quartz ($\rho_q = 2650$ kg m^{-3}) and feldspar ($\rho_f = 2600$ kg m^{-3}). The volumetric fractions of solids are f_q and f_f with $f_q + f_f = 1$.

Tasks

1. Write an expression for bulk density:
2. Compute ρ_b for: (i) $\phi = 0.30$, air-filled pores ($\rho_{\text{fluid}} \approx 1$), (ii) $\phi = 0.30$, water-filled pores ($\rho_{\text{fluid}} = 1000$), (iii) $\phi = 0.30$, brine-filled pores ($\rho_{\text{fluid}} = 1200$).
3. Repeat for $\phi = 0.10$ (burial compaction).
4. Explain physically:
    - Why does ρ_b increase with reduced ϕ?
    - Which fluid substitution changes ρ_b the most?